

Milvus + Docker in RAG Project - Summary Documentation

Why We Need a Vector Database

In a RAG system, PDFs are converted into chunks and embeddings. Traditional databases are not optimized for semantic similarity search, so we use a vector database.

Why We Chose Milvus

Milvus is an open-source vector database designed for AI applications, semantic search, recommendation systems, and Retrieval-Augmented Generation (RAG). It supports fast approximate nearest neighbor search algorithms such as HNSW and IVF.

Benefits of Milvus

- Fast vector similarity search
- Scales to millions and billions of vectors
- Production-ready and open source
- Excellent Python integration
- Supports metadata filtering and advanced indexing.

Why We Used Docker

Milvus requires multiple services and dependencies. Docker provides an isolated, reproducible environment and allows us to start Milvus with a single command.

Milvus Deployment Used in Our Project

We used Milvus Standalone with Docker image: milvusdb/milvus:v2.5.0.

Important Docker Commands

`docker compose up -d` → Start Milvus
`docker ps` → Check running containers
`docker compose down` → Stop containers
`docker logs milvus` → View logs

Ports Used

19530 → Main Milvus API port used by Python.
9091 → Management and health-check port.

Collection Schema

Collection Name: documents

Fields:

- id (INT64 Primary Key)
- text (Original Chunk)
- embedding (FLOAT_VECTOR, dimension=1024).

Ingestion Pipeline

PDF → Loader → Chunking → Embedding Model (BGE Large) → Milvus Storage.

Retrieval Pipeline

User Question → Query Embedding → Milvus Similarity Search → Top-K Chunks → Context.

Complete RAG Architecture

PDF → Chunking → Embedding → Milvus → Retrieval → Context → LLM (Phi-3) → Final Answer.

Interview Summary

Why Milvus? Because it is optimized for vector similarity search.

Why Docker? To simplify setup and provide an isolated environment.

Why Embeddings? To represent semantic meaning as vectors.

Why LLM? Retrieval finds information; the LLM generates natural-language answers from that information.